Hydrological & Environmental Monitoring of the Blessington Lakes using SWOT and Sentinel-1 Data

Assessing Lake Elevation Variability for Water Availability and Surface Reflectivity Variations Associated with Potential Algal Presence in the Blessington Lakes using Sentinel-2 and SWOT Data.

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Abstract

This project looks at how the Blessington Lakes (Poulaphouca Reservoir) change over time, focusing on both lake elevation and algal presence using Sentinel-2 and SWOT satellite data. The reservoir is one of the most important water sources in Ireland, supplying over half of Dublin's drinking water, so understanding how its water levels vary throughout the year (weather and human intervention dependant) is essential for monitoring water availability and overall environmental health. Using SWOT data, I examined changes in lake elevation and compared them with rainfall and water level records from local stations. In addition, Sentinel-2 optical images were used to explore whether there were signs of algal activity during low-water periods. The early results show that water levels in the lake fluctuate seasonally and respond closely to rainfall patterns. Changes in occasional higher reflectivity in Sentinel-2 imagery might point to surface greening or algae. These findings highlight how combining radar and optical satellite data can help monitor inland water bodies like the Blessington Lakes, even with some limits in data resolution and consistency.

Highlights

- Investigates seasonal lake elevation changes in the Blessington Lakes, a major drinking water source for Dublin.
- Uses a combination of SWOT water surface elevation data and Sentinel-2 SAR imagery to monitor water level and purity variability.
- Compares satellite-derived results with rainfall and hydrometric data to assess relationships with water availability.
- Demonstrates the potential of open-access radar and optical data for ongoing monitoring of Irish inland water bodies.
- Notes limitations in spatial resolution and radar coherence over open water, suggesting areas for future improvement.

Introduction

The Blessington Lakes (also known as the Poulaphouca Reservoir) in County Wicklow are one of Ireland's most important freshwater resources. These lakes currently provide an estimated 53% of Dublin's drinkable tap water. Manmade specifically for this purpose. The lakes were created between the late 1930s to the early 1940s through the damming of the infamous River Liffey in the valley area between the Valleymount and Blessington hills. This was completed through the eviction of 74 local families from their homes and lands. These families received minimal compensation of the equivalent of €30,000 in today's monetary value. The lakes and reservoir also currently support hydroelectric power generation and local recreation. As a managed system, it experiences ongoing fluctuations in water levels due to rainfall, inflow from upland catchments, and water abstraction for supply. During the summer of 2024, an unusually dry summer period revealed large areas of exposed shoreline, old home and church structures and furniture which was buried long ago. This drought raised concerns about how such seasonal and climatic changes may impact the lakes' initial purpose and, also ignited ideas and questions within me of whether the lake can be measured and monitored using satellite remote sensing. Monitoring lake elevation and surface conditions is essential for understanding how reservoirs respond to changing rainfall patterns and human water use in day-to-day life. Traditional ground-based hydrometric gauges provide point-based measurements, but unfortunately; do not capture the full spatial variability of the reservoir or surrounding catchment. Remote sensing, particularly using radar and altimetry satellites, offers a consistent and wide-area view that can complement local monitoring efforts.

The Sentinel-2 mission, part of the European Space Agency's Copernicus programme, is an optical multispectral system that provides 10 to 60 m resolution imagery across 13 spectral bands with a five-day revisit cycle. It is visible and infrared bands are well suited for monitoring water quality, vegetation, turbidity, algal blooms, and shoreline position. Changes in spectral reflectance can highlight suspended sediment, coloured dissolved organic matter, and floating or submerged vegetation, allowing detailed mapping of water boundaries and surface conditions in inland lakes and reservoirs. The Surface Water and Ocean Topography (SWOT) satellite, launched by NASA and CNES in 2022, compliments Sentinel-2 by measuring water surface elevation directly with Ka-band radar interferometry, enabling combined analysis of water extent, surface properties, and elevation dynamics. In addition, to elevation levels, water quality is an important consideration for reservoirs supplying drinking water. Although the Blessington Lakes are not usually associated with algal blooms, periods of high temperature and low water depth can contribute to biological growth. Optical imagery from the Sentinel-2 mission allows the calculation of vegetation and chlorophyll indices, which can serve as proxies for detecting potential algal activity. Combining this with radar-based data offers an insightful view of both the physical and environmental conditions of the reservoir.

This study aims to explore how open-access radar and optical satellite data can be used to monitor hydrological and environmental changes in the Blessington Lakes. The main objectives are to:

1. Assess the lake elevation levels and changes using SWOT data
2. Evaluate water level data with existing rainfall and in-situ water level measurements.
3. Investigate Sentinel-2 optical imagery for signs of surface greening or potential algal presence during low-water periods.
4. Discuss the strengths and limitations of combining these datasets for reservoir monitoring in the Irish context.

By focusing primarily on elevation analysis and secondarily on potential algae detection, this project demonstrates how satellite remote sensing can enhance understanding of inland water dynamics. The integration of SWOT and Sentinel-1 data provides insight into the relationship between climate variability, water availability, and surface conditions in one of Ireland's most critical freshwater systems.

Methods

Research Study Area

This research focuses on the Blessington Lakes, Poulaphouca Reservoir, situated in Northwestern County Wicklow, on the Irish eastern coast, located at approximately 53.13° N, -6.53° E. The reservoir covers an area of approximately 16 km². It plays a vital role in providing roughly half of Dublin's drinking water and contributes to hydroelectric power generation. The surrounding landscape includes peatland catchments, forested slopes, and rural farmland, which influence water inflow and quality. The region has a temperate maritime climate, with average annual rainfall of just over 1,200mm. Seasonal variations in rainfall occasionally result in noticeable changes in the reservoir's water level and exposed shoreline.

Datasets

- **SWOT Water Surface Elevation Data**
  Water surface elevation measurements were obtained from the Surface Water and Ocean Topography (SWOT) mission, a joint NASA and CNES satellite launched in late 2022. SWOT's Ka-band radar interferometer provides water surface elevation at approximately 50-100 m spatial resolution and a 21-day revisit period. Level 3 Water Surface Elevation (WSE) products (Version 1.1) were downloaded directly from NASA's PO.DAAC data portal. The timeframes analysed and downloaded during this research include all available swathe data recorded from the January 1st of 2024, until December 4th of 2025.

- **Sentinel-2 Optical Data**
  Sentinel-2 MSI Level-2A imagery was used to examine possible optical indicators of algal activity. Bands 4 (red) and 8 (near-infrared) were used to calculate the Normalized Difference Vegetation Index (NDVI) as a proxy for surface greenness or bloom presence. Images from July-September 2024 and June-August 2025, these are two timeframes when algal growth was most likely, were selected to match the dry-season SWOT acquisitions.

- **Ancillary Data**
  Daily rainfall records were acquired from nearby local Met Éireann weather stations (Valleymount, Brittas and Glenbridge Lodge) and National weather and metrology stations (Casement Aerodrome & Oak Park). In-situ water level data for the reservoir were attempted to be obtained from the Environmental Protection Agency (EPA) hydrometric network. These datasets were used to validate, interpret and compare to the acquired satellite-derived results.

Data Pre-Processing

SWOT Data

SWOT Level-2 HR LakeSP files were subset to the lake area using QGIS 3.40.12. Water surface elevation values were averaged across the lake for each acquisition and exported to a CSV for time-series comparison with rainfall and EPA water level records.

Analysis
Satellite-derived elevation datasets were compared against ground rainfall and water-level in-situ observations to investigate relationships between hydrological variability and satellite measurements. Time-series plots were produced to visualise seasonal patterns. Pearson correlation coefficients were calculated between SWOT, EPA, and rainfall data. For optical analysis, mean NDVI values from Sentinel-2 were plotted to explore potential optical greenness.

Reproducibility
All data used in this project are open-access and publicly available through ESA Copernicus and NASA PO.DAAC. Processing was carried out using ESA SNAP 9.0, QGIS 3.40, and Python 3.11 for data handling and plotting. The workflow, parameters, and coordinate system were documented to ensure reproducibility of results.

Results

The integration of the SWOT and Sentinel-2 data provided complementary perspectives on the hydrological and environmental behaviour of the Blessington Lakes. The SWOT-derived elevation time series captured clear seasonal fluctuations that matched rainfall patterns and ground-based water-level data. Meanwhile, Sentinel-2 optical imagery suggested somewhat limited, but suggestive and observable changes in surface reflectivity during low-water months. Together, these results demonstrate the ability of multi-sensor satellite data to track both hydrological variation and surface condition changes in Irish reservoirs. While the temporal coverage of SWOT and the coherence limits of Sentinel-2 impose constraints, the combination of both datasets produces a coherent picture of the reservoir's seasonal dynamics.

Results 1; SWOT Water Elevation.

Data Acquisition

All the data was downloaded directly from PO.DAAC using the 'iCrag/Ez-InSar/Hydro-Module' created for this module found on https://github.com/alexisInSAR/EZ-InSAR by the module coordinator; Alexis Hrysiewicz. The environment on was accessed via Ubuntu Linux operating system and downloaded the data directly to a dedicated data folder (see Fig below). The commands used for data acquisition are screenshotted and provided below, in addition to a pastable set found in the 'Appendix' section.

```
(base) orlaithnidhuill@Orlaith:~$ # 1. Activate EZ-InSAR HydroOcean environment
conda activate EZ-InSAR-3

# 2. Create a working directory for Blessington SWOT data
mkdir -p ~/SWOT_Blessington
cd ~/SWOT_Blessington

# 3. DOWNLOAD SWOT L2 LAKE SP PRODUCTS FOR BLESSINGTON

ezinsar hydroocean download \
    -s SWOT \
    -b -6.618940,53.068340,-6.412946,53.210033 \
    --download \
    -u 'EarthData_Username' \ #This needs to be edited
    -p 'EarthData_Password' #This too

# 4. EXTRACT SWOT LAKE DATA TO GEOJSON
ezinsar hydroocean extractSWOT \
    -i ./ \
    -b -6.618940,53.068340,-6.412946,53.210033 \
    -o blessington_raw.geojson \
    -m Lake

# 5. FILTER BY THE ACTUAL BLESSINGTON LAKE ID

python3 << EOF
import geopandas as gpd

# Load extracted SWOT LakeSP GeoJSON
g = gpd.read_file('blessington_raw.geojson')

# Filter for the Blessington/Poulaphouca lake_id
EOFnt("Filtered Blessington file written to Blessington_LakeID.geojson")
```

Figure 1. Code used in Ubuntu Terminal to obtain SWOT data and download folders to dedicated Data folder.

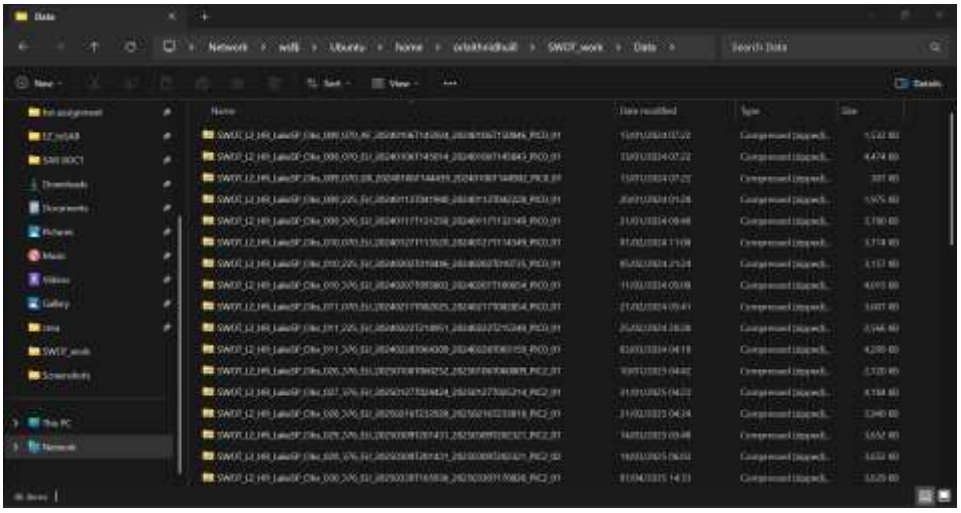


Figure 2. Data folder containing all SWOT data downloaded from PO.DAAC.

QGIS Results

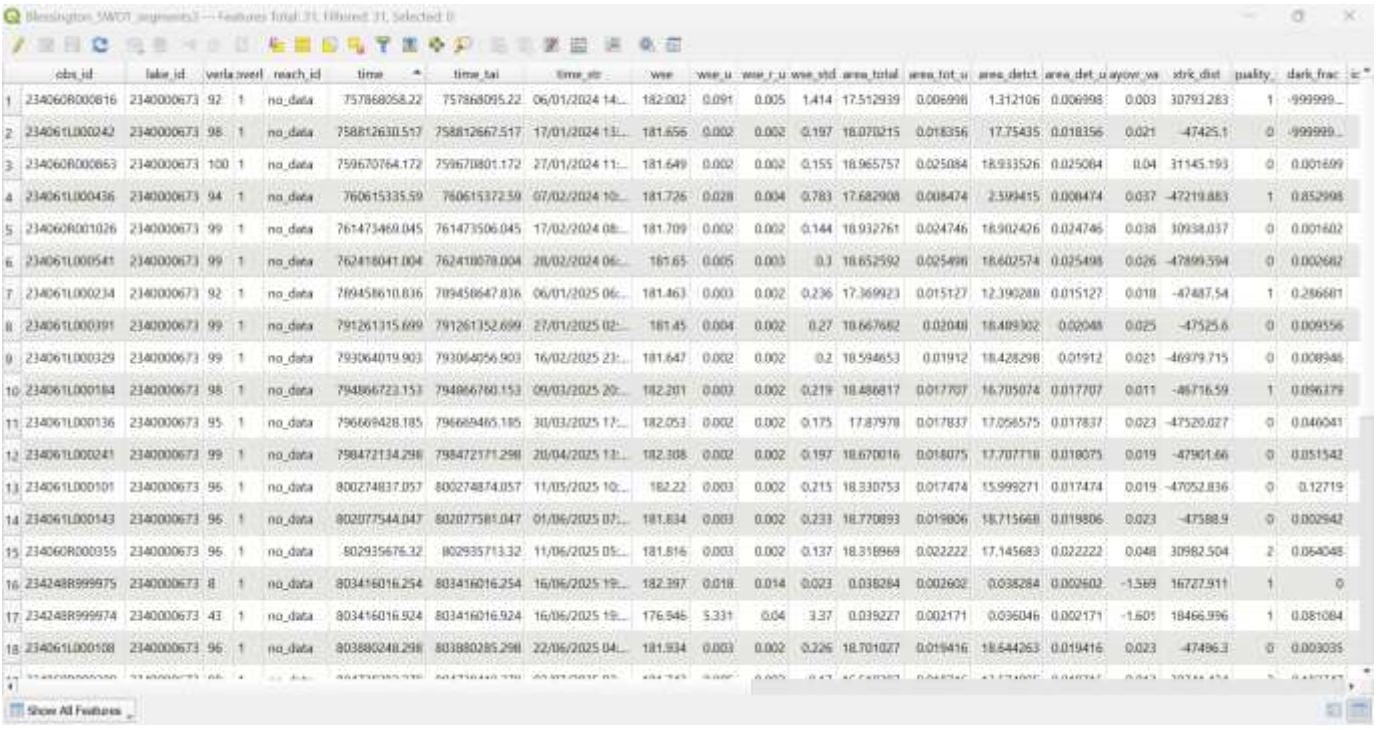
To visualise the SWOT outputs spatially and to obtain real-world data such as WSE and similar data, I imported Ubuntu downloaded LakeSP files into QGIS and clipped each polygon layer to the Blessington Lakes boundary so only observations directly overlapping Poulaphouca were retained. Once the layers were loaded, the combined

attribute table became the main way of checking what each pass represented. This table includes the lake_id, the WSE values, the uncertainty fields, the exact acquisition date and time, and several additional observation descriptors such as overlap count and detected area. These titles are explained in more depth in the 'Appendix', where I have colour-coded them to separate elevation-related fields, identification fields, and date/time information, making it clearer which parameters mattered for this analysis.

After confirming the fields I needed, I created three separate layers for comparison:

1. All overlapping SWOT polygons, displayed in bright red, to show every pass that intersected the lake. *[Figure 5]*
2. Maximum WSE polygon, symbolised in light red. *[Figure 7]*
3. Minimum WSE polygon, symbolised in green. *[Figure 6]*
4. A combined overlap of both the WSE min & max to highlight observational differences *[Figure 8]*

To do this, I filtered the attribute table by WSE values, exported the max and min subsets as new shapefiles, and styled them using simple graduated colours so the differences would stand out clearly. When the three layers were visualised together, the spatial pattern was obvious: the red layer dominates the reservoir surface, covering nearly the entire lake across multiple passes, while the green polygons representing minimum WSE are much more limited in extent and only appear in small, isolated patches. This matched what I already saw in the elevation time-series plots and helped verify that the unusually low WSE outliers were not consistent across the swaths. I also used the attribute table to cross-check the observation times for each polygon, comparing them against the corrected 21-day acquisition cycle. This confirmed which passes belonged to overlapping swaths and which ones came from the problematic June 16th recordings *(Figure 4)*. Having the spatial context in QGIS made it clear how the satellite had viewed the lake from different angles, especially during periods where the orbit geometry produced multiple swaths over the same area. Overall, the QGIS results were needed for validating the SWOT time-series visually and for understanding how the lake was physically captured across different passes. The combination of the attribute table filtered WSE subsets, and the colour-coded breakdown in the Appendix helped keep track of how each observation related to elevation, timing, and swath overlap.



	obs_id	lake_id	varla	sweil	reach_id	time	time_tai	time_str	wse	wse_u	wse_r_u	wse_std	area_total	area_tot_u	area_detct	area_det_u	ayow_wa	xtrk_dist	quality	dark_frac
1	234060R000816	2340000673	92	1	no_data	757868058.22	757868095.22	06/01/2024 14:...	182.002	0.091	0.005	1.414	17.512939	0.006998	1.312106	0.006998	0.003	30793.283	1	-999999...
2	234061L000242	2340000673	98	1	no_data	758812630.517	758812667.517	17/01/2024 13:...	181.656	0.002	0.002	0.197	18.070215	0.018356	17.75435	0.018356	0.021	-47425.1	0	-999999...
3	234060R000863	2340000673	100	1	no_data	759670764.172	759670801.172	27/01/2024 11:...	181.649	0.002	0.002	0.155	18.965757	0.025084	18.933526	0.025084	0.04	31145.193	0	0.001699
4	234061L000436	2340000673	94	1	no_data	760615335.59	760615372.59	07/02/2024 10:...	181.726	0.028	0.004	0.783	17.682908	0.008474	2.599415	0.008474	0.037	-47219.883	1	0.852998
5	234060R001026	2340000673	99	1	no_data	761473469.045	761473506.045	17/02/2024 08:...	181.709	0.002	0.002	0.144	18.932761	0.024746	18.902426	0.024746	0.038	30938.037	0	0.001602
6	234061L000541	2340000673	99	1	no_data	762418041.004	762418078.004	28/02/2024 06:...	181.65	0.005	0.003	0.3	18.652592	0.025498	18.602574	0.025498	0.026	-47899.594	0	0.002682
7	234061L000234	2340000673	92	1	no_data	789458610.836	789458647.836	06/01/2025 06:...	181.463	0.003	0.002	0.236	17.369923	0.015127	12.390288	0.015127	0.018	-47487.54	1	0.286681
8	234061L000391	2340000673	99	1	no_data	791261315.699	791261352.699	27/01/2025 02:...	181.45	0.004	0.002	0.27	18.667662	0.02048	18.489302	0.02048	0.025	-47525.6	0	0.009556
9	234061L000329	2340000673	99	1	no_data	793064019.903	793064056.903	16/02/2025 23:...	181.647	0.002	0.002	0.2	18.594653	0.01912	18.428298	0.01912	0.021	-46979.715	0	0.008946
10	234061L000184	2340000673	98	1	no_data	794866723.153	794866760.153	09/03/2025 20:...	182.201	0.003	0.002	0.219	18.486817	0.017707	16.705074	0.017707	0.011	-46716.59	1	0.096379
11	234061L000136	2340000673	95	1	no_data	796669428.185	796669465.185	30/03/2025 17:...	182.053	0.002	0.002	0.175	17.87978	0.017837	17.056575	0.017837	0.023	-47520.027	0	0.046041
12	234061L000241	2340000673	99	1	no_data	798472134.298	798472171.298	20/04/2025 13:...	182.308	0.002	0.002	0.197	18.670016	0.018075	17.707718	0.018075	0.019	-47901.66	0	0.051542
13	234061L000101	2340000673	96	1	no_data	800274837.057	800274874.057	11/05/2025 10:...	182.22	0.003	0.002	0.215	18.330753	0.017474	15.999271	0.017474	0.019	-47052.836	0	0.12719
14	234061L000143	2340000673	96	1	no_data	802077544.047	802077581.047	01/06/2025 07:...	181.834	0.003	0.002	0.233	18.770893	0.019806	18.715668	0.019806	0.023	-47588.9	0	0.002942
15	234060R000355	2340000673	96	1	no_data	802935676.32	802935713.32	11/06/2025 05:...	181.816	0.003	0.002	0.137	18.318969	0.022222	17.145683	0.022222	0.048	30982.504	2	0.064048
16	234248R999975	2340000673	8	1	no_data	803416016.254	803416016.254	16/06/2025 19:...	182.397	0.018	0.014	0.023	0.038284	0.002602	0.038284	0.002602	-1.569	16727.911	1	0
17	234248R999974	2340000673	43	1	no_data	803416016.924	803416016.924	16/06/2025 19:...	176.946	5.331	0.04	3.37	0.039227	0.002171	0.036046	0.002171	-1.601	18466.996	1	0.081084
18	234061L000108	2340000673	96	1	no_data	803880248.298	803880285.298	22/06/2025 04:...	181.934	0.003	0.002	0.226	18.701027	0.019416	18.644263	0.019416	0.023	-47486.3	0	0.003035

Figure 3. Data table for downloaded SWOT datasets including date timestamps, dates, lake ID and WSE information.

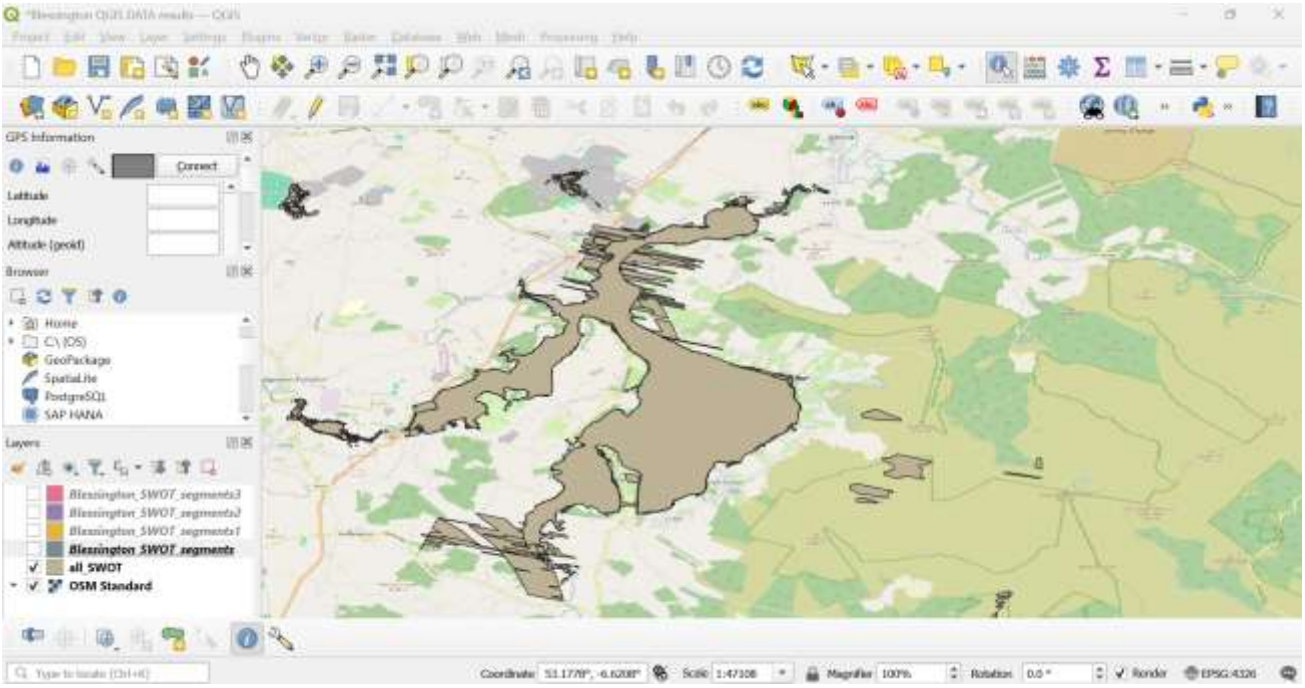


Figure 4. All downloaded SWOT data visualised using QGIS with OSM standard map for location referencing.

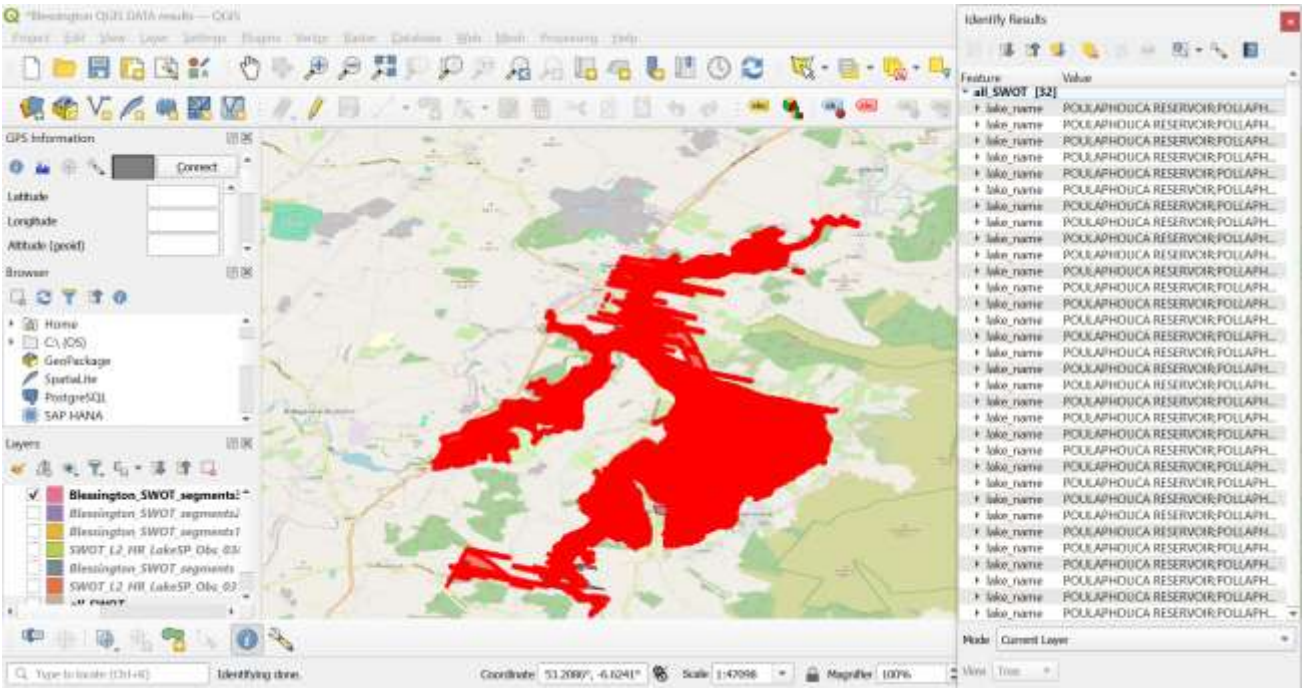


Figure 5. Corrected SWOT data visualisation for exact focus on the lake area only (Minimal Error Interference visual stack for 32 SWOT datasets)

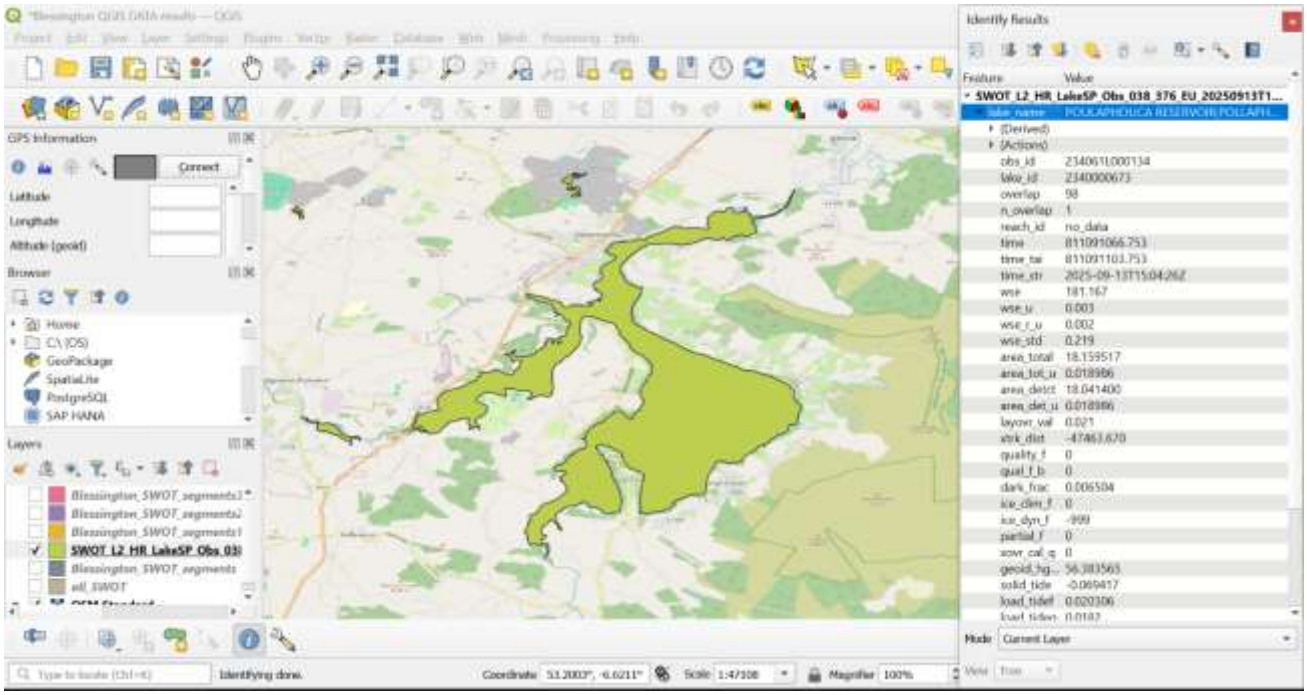


Figure 6. Visualisation and Attribute Data of the 'WSE Min' on the 13th of September 2025.

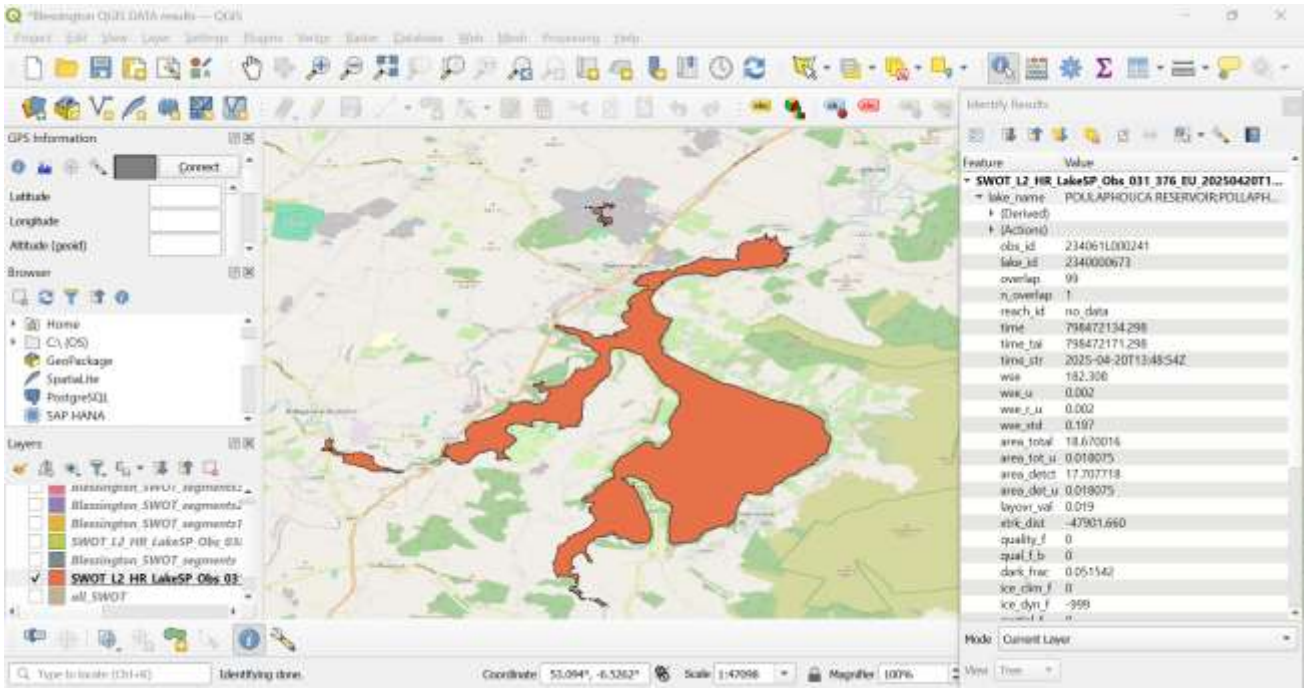


Figure 7.Visualisation with Attribute data results of the 'WSE Max' on 20th of April 2025.

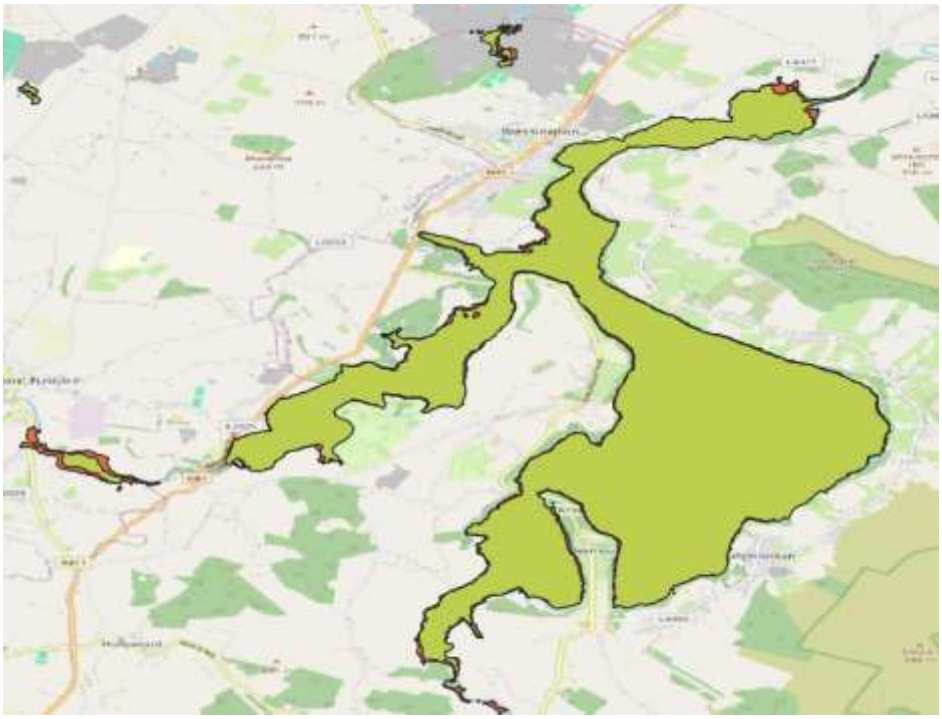


Figure 8. Overlapped max/min comparison. It can be clearly noted that the 'max' data obviously has slightly more surface area inline with the slightly higher elevation levels of the water during this data swathe.

The SWOT-derived WSE values showed the lake's hydrological changes really clearly, even though the data wasn't available consistently across the whole study period. Only a small set of passes appeared for early 2024, which makes sense because SWOT was still in its Cal/Val phase at that point and inland water products were being released on and off as they were tested. After that, there's basically a big gap right through to the end of 2024, which lines up with the mission shifting into its fully validated science phase before regular global LakeSP products were made available.

From January 2025 onwards the dataset actually becomes useful. The satellite finally settles into stable operations and starts producing proper LakeSP observations over Ireland. Once I exported the QGIS attribute tables to CSV and pulled them into both Excel and Python for plotting, the seasonal pattern stood out straight away. The WSE gradually drops through spring and summer, bottoming out in mid-August during the dry spell, and then climbs back up sharply after the heavy rainfall in October and November.

Rainfall Data Comparison (Early 2024 & 2025)

To understand how the water-level changes related to rainfall, I compared the SWOT WSE time series with both local and national Met Éireann datasets. For the early 2024 period, I used the nearest available daily rainfall stations; Ballymore Eustace, Brittas, and Glenbridge Lodge, which together form a triangulated set around the Blessington Lakes (see relative locations in Figures below). These were the closest operational stations with data during SWOT's January to February 2024 Cal/Val window. When plotted, the daily rainfall pattern from these three sites showed a run of relatively low but frequent rainfall events, which matches the small WSE fluctuations seen in the early SWOT passes.

For the full 2025 analysis, I expanded the comparison by including the national monthly datasets from Oak Park (Carlow) and Casement Aerodrome (South Dublin), as the closer stations did not have complete year-round records. These larger stations provided a more stable monthly rainfall record. Once the data was plotted together, the pattern became really clear: rainfall steadily dropped through spring and summer 2025, reaching its lowest point around July to August, which directly aligns with the minimum WSE values extracted from the corrected SWOT dataset. After that, both stations show a sharp increase in rainfall during October and November, which mirrors the rapid rebound in lake elevation captured by SWOT during the same months.

Overall, comparing the local daily stations for early 2024 and the national monthly stations for 2025 helped confirm the relationship between rainfall and lake behaviour, even when taking into account that the lake elevation is independently managed by ESB. Even though the datasets differ in scale and resolution, they support the same hydrological story; low rainfall drives the summer drawdown, and the autumn rainfall pulses are responsible for the quick recovery in water levels. When I compared this with the EPA hydrometric data, the trends matched really well, even though the absolute heights are slightly different because of how SWOT averages values across the lake surface. Overall, even with the early data gaps and the 21-day revisit, SWOT captured the expected drawdown-recharge cycle of the Blessington Lakes. The patterns tie in with rainfall, lake exposure, and storage changes, showing that once the mission reached full data maturity in 2025, it became a reliable tool for tracking these elevation shifts over time.

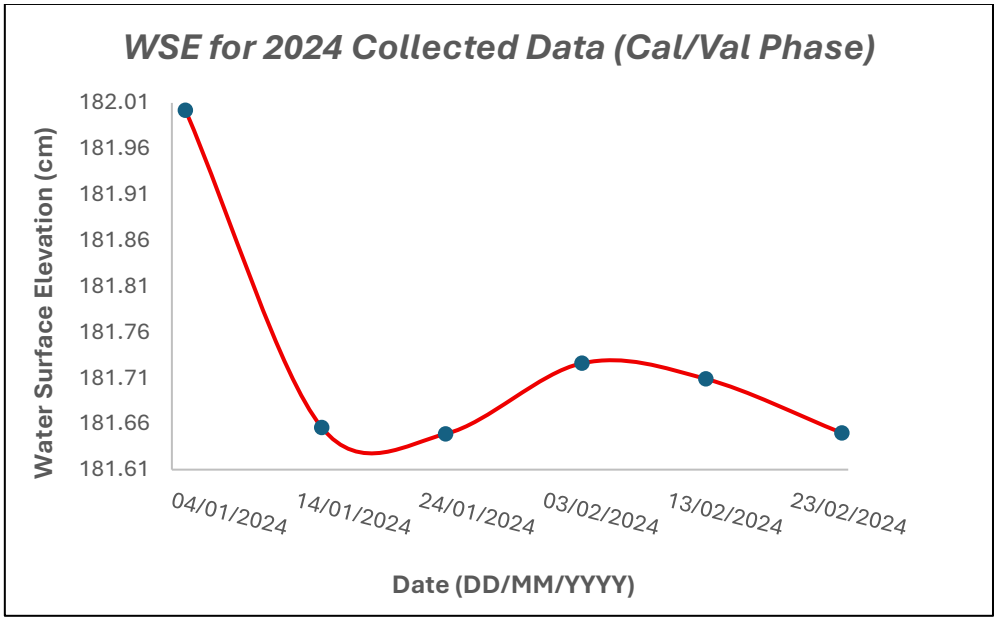


Figure 9. Water Surface Elevation for 2024 SWOT passes.
Note; there is limited data available as this was during the 'Calibration Validation' Phase.

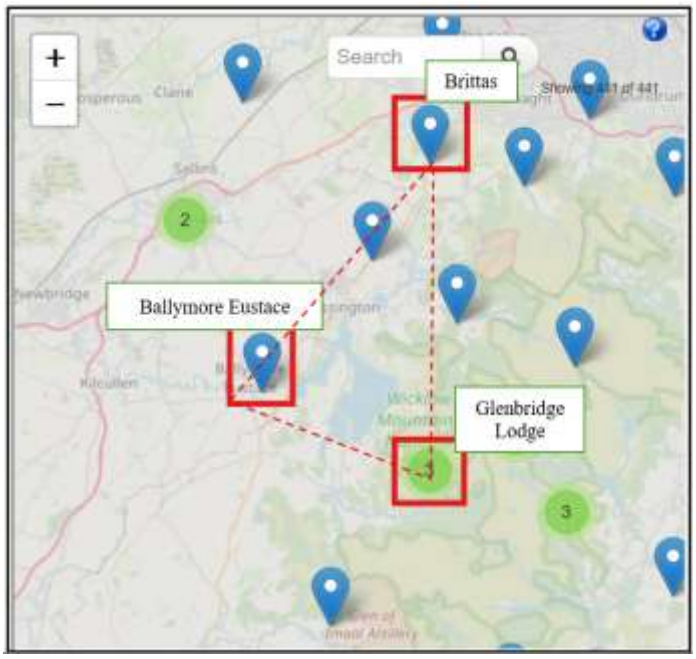


Figure 10. Local weather station locations relative to the Blessington Lakes. [Note; there was no data available for closer stations such as Blessington (North of Lake), Ballinatona (West of Lake) or Glen of Imaal (South of Lake). Data Source; https://www.met.ie/climate/available-data/historical-data].

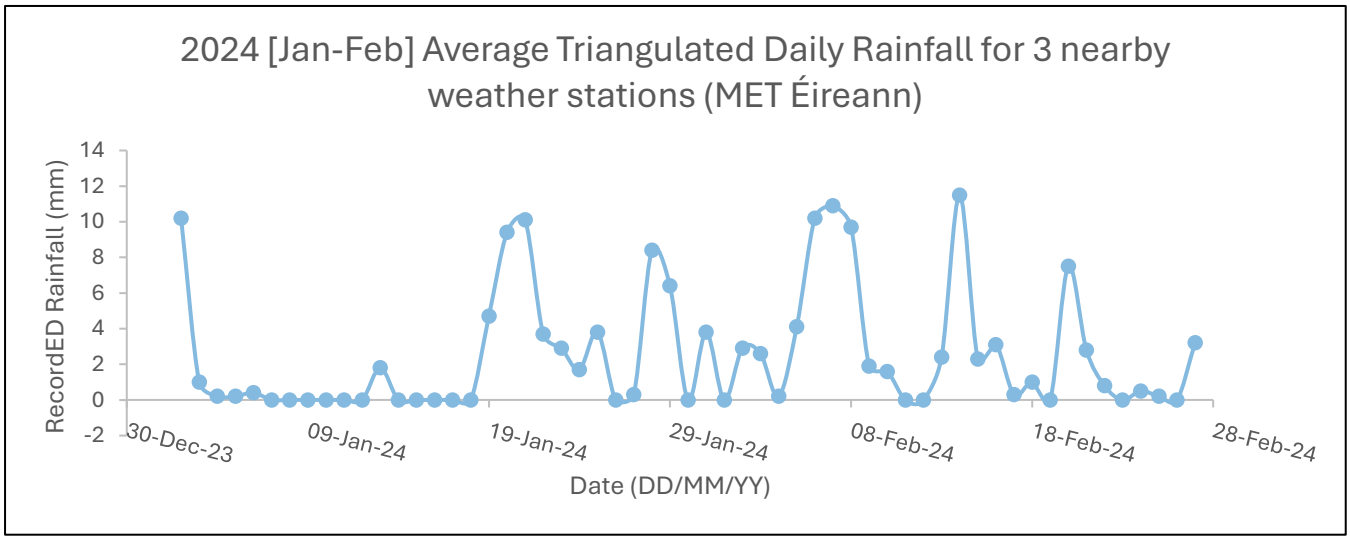


Figure 11. Average Recorded Daily Rainfall for Triangulated local weather stations within nearby region around Blessington Lakes.
[Weather Station Data; https://www.met.ie/climate/available-data/historical-data]

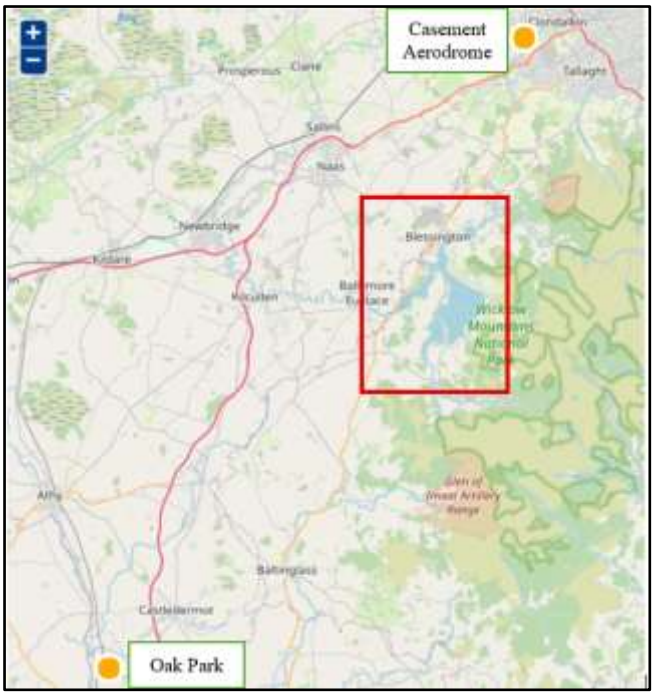


Figure 12. Locations relative to Blessington Lakes of nearby National weather stations which provide monthly & more up-to-date recorded precipitation data of rainfall. https://www.met.ie/climate/available-data/monthly-data

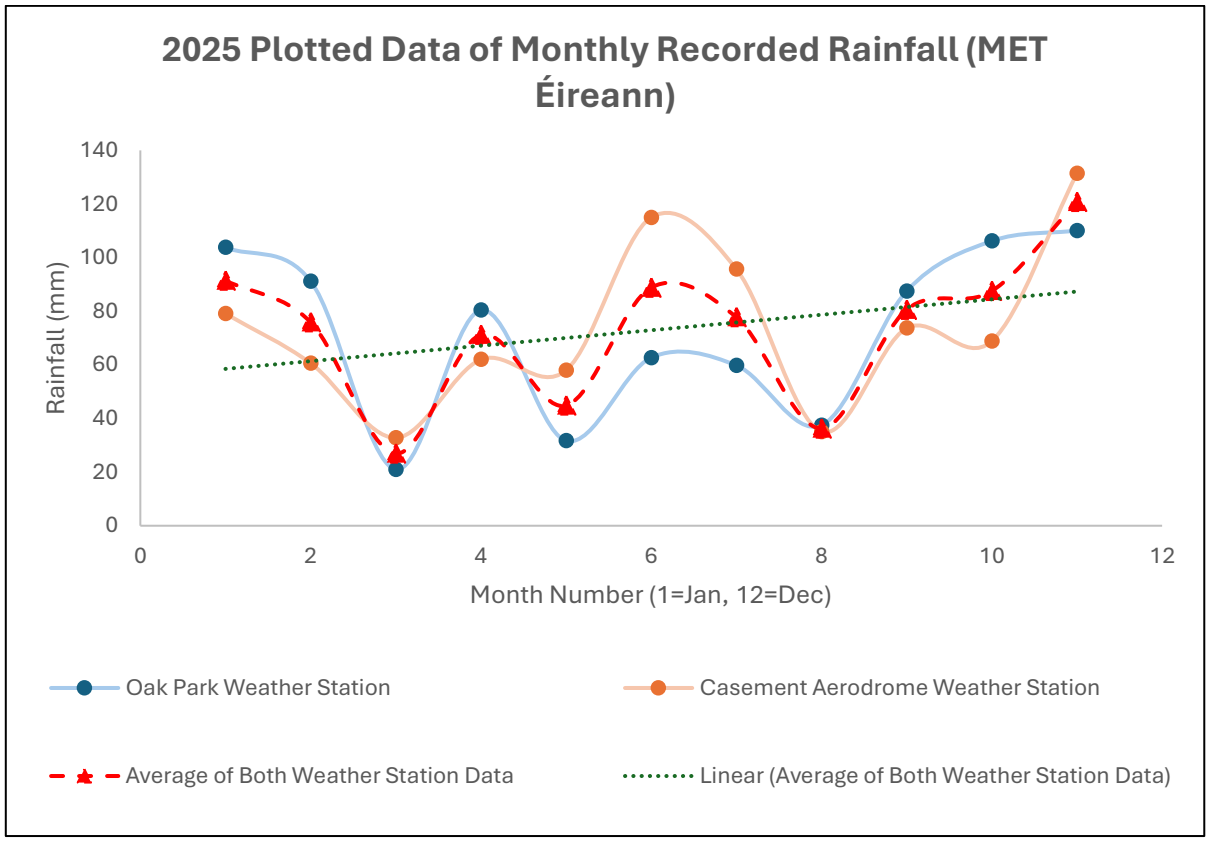


Figure 13. Plotted recorded rainfall data from both 'Oak Park' & 'Casement Aerodrome' National weather stations. [https://www.met.ie/climate/available-data/monthly-data]

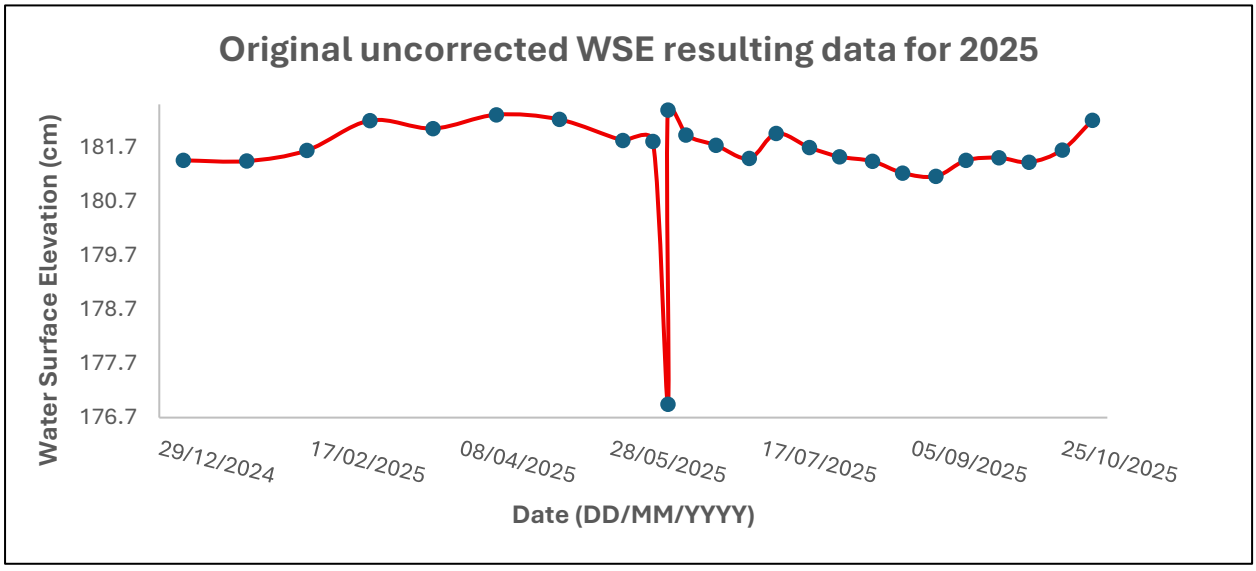


Figure 14. Uncorrected plotted Water Surface Elevation (WSE) from SWOT data for the region of interest; Blessington Lakes (Poulaphouca Reservoir). Note; The two contradictory SWOT observations on 16 June 2025, and the more frequent ~15-day sampling that follows, are due to the lake sitting inside overlapping swaths after an orbit phase shift. This can create extra passes within a 21 day cycle and is a normal feature of SWOT's wide KaRIn coverage (Morrow et al., 2023; Altenau et al., 2024; Lichtman et al., 2025) [https://podaac.jpl.nasa.gov/SWOT]

It seems like a perfect set of data, only there is too much information provided by the SWOT data, the data downloaded after June 1st seems to be unreliable as the swathe becomes more frequent than 21days, and the next pass should've been on June 23rd, instead there are two intermittent sets both on June 16th both with completely different WSE results, one at a very notable minimum outlier of 176.946cm and the other providing a maximum elevation value of 182.397cm. I have no other explanation for these values other than that they may have been produced in due to swathe overlapping for different geometrical angles as they are completely opposing min/max values. Based on this assumption, I have altered the original dataset to exclude the errored recorded data and the infrequent extra recorded data so that the results can be examined with eased for a clear comparison of the elevation levels across 2025.

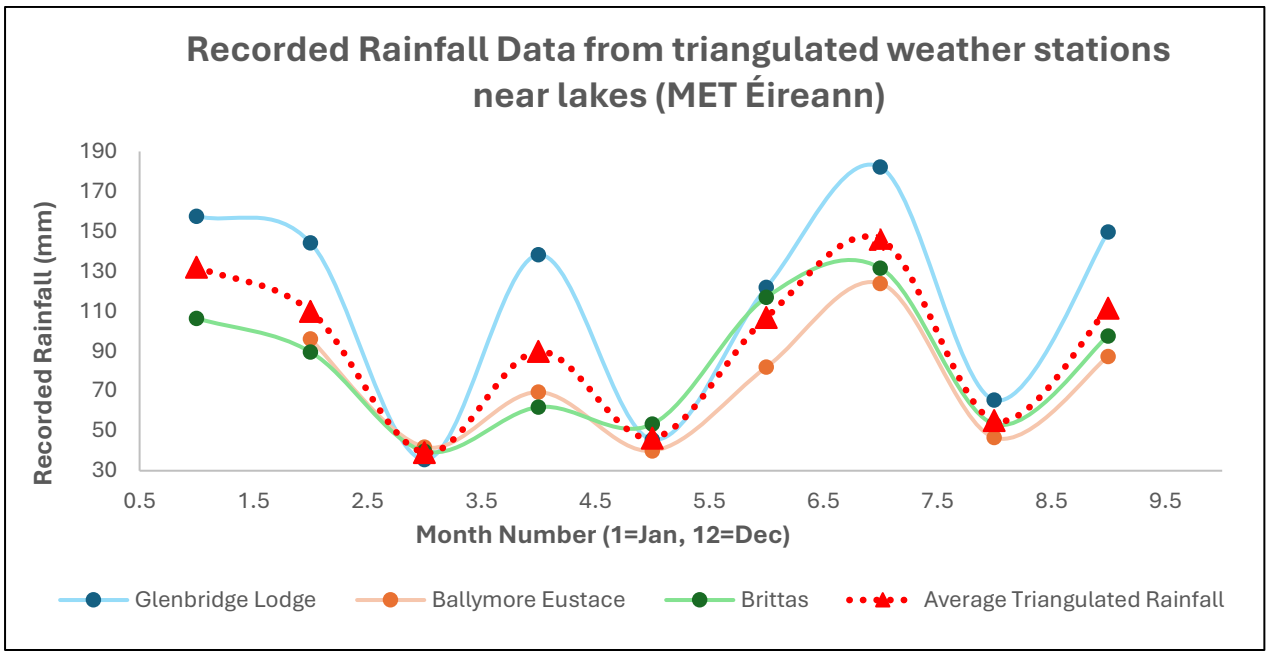


Figure 15. 2025 monthly triangulated data for rainfall with average clearly stated (red).

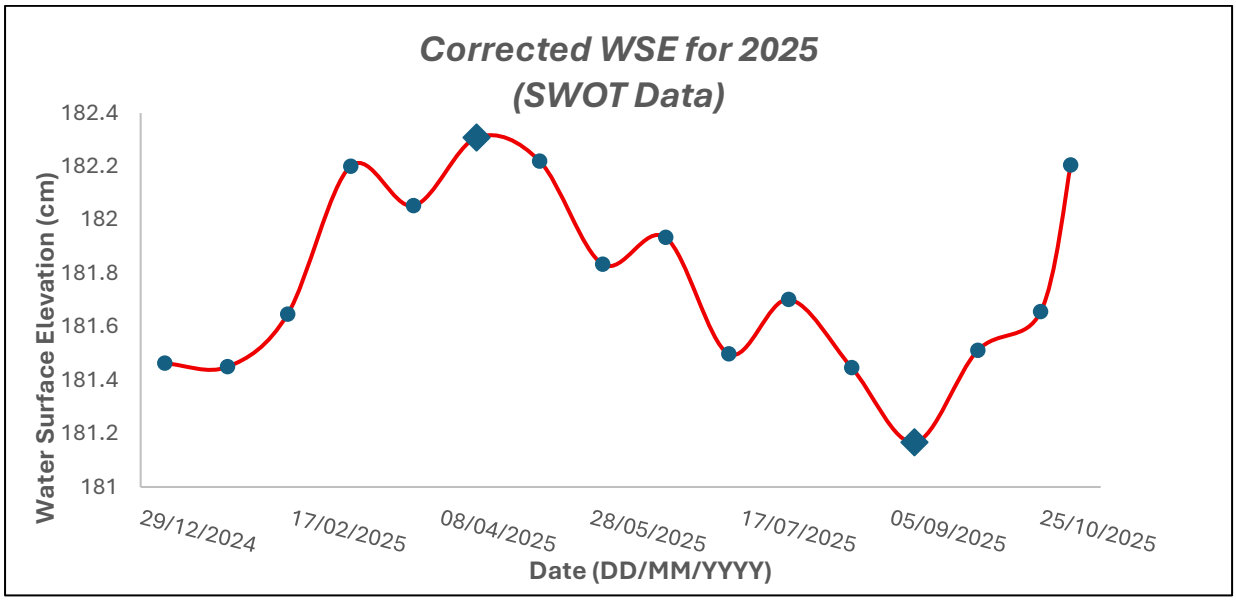


Figure 16. Corrected 2025 SWOT Data with consistent 21-day cycle results and removal of error min/max recorded data from June 16th.

I have attempted to obtain in-situ water level data for the lake from the EPA Hydronet database, but no elevation records were available for this site as the lake is owned and managed by The Electric Supply Board (ESB), its hydrometric measurements are not part of the public EPA network, which typically provides water level, flow, and rainfall records for monitored catchments. See below Figures for further clarification of data available. Unfortunately, access to non-public stations must be requested directly through Hydronet. I have submitted a request for these records, but no response has been received to date. I will update this report once/if received.



Figure 17. Screenshots of available EPA data for nearby Kilcullen Stream Lake elevation data (right) [https://epawebapp.epa.ie/hydronet]

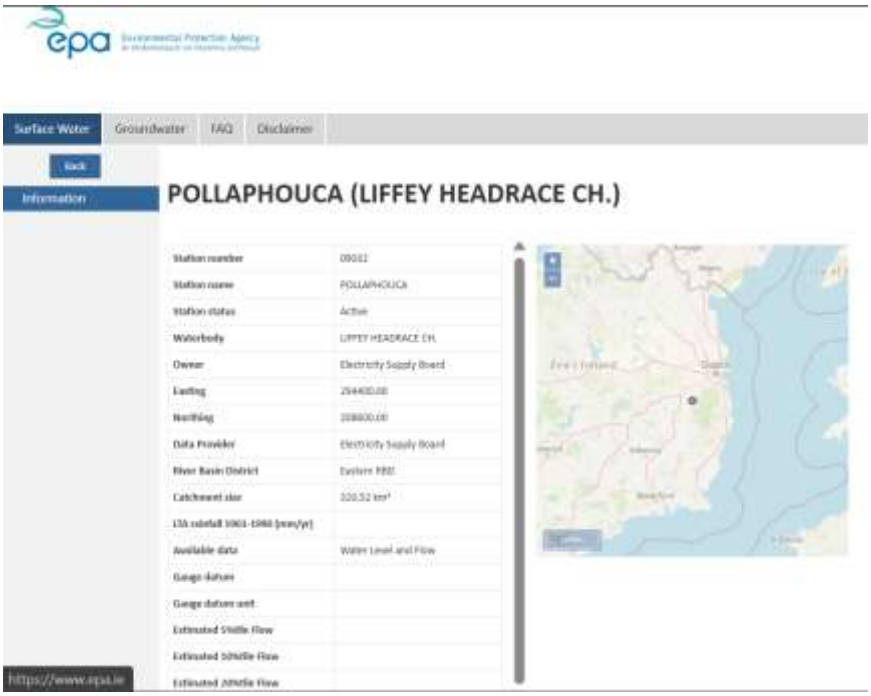


Figure 18. Limited available data for Poulaphouca Reservoir (Local to Blessington Lakes). All further available data must be requested for access. They have yet to send an update on this requested information.*

Results 2; Algal Detection using Sentinel-2 Imagery.

The Sentinel-2 NDVI imagery for August 2025 was used to assess potential algal activity. I began the analysis by looking at NDVI and NDWI outputs in the Copernicus interactive viewer (see Figs below) to identify any obvious surface greening or changes in water reflectance. I had originally intended to analyse the dry spell of August 2024 when lake levels were at their lowest and conditions were warm and sunny, which would have been an ideal window to observe algal growth. However, clear cloud-free Sentinel-2 scenes from 2024 were difficult to obtain in Copernicus, and I was also limited by time and storage capacity. As a result, the analysis focused on August 2025 instead, which also represented a dry and warm period suitable for indicative algal assessment. The resulting time-series optical imagery and filtered NDVI outputs showed small but noticeable increases in average NDVI values along shallow nearshore areas during the lowest water-level stages. These patterns suggest short-lived surface greening that may be linked to algal material, floating vegetation, or exposed sediment. No widespread or persistent algal bloom was detected, but the timing of these localised anomalies corresponds with the dry and warm conditions recorded during the same period.

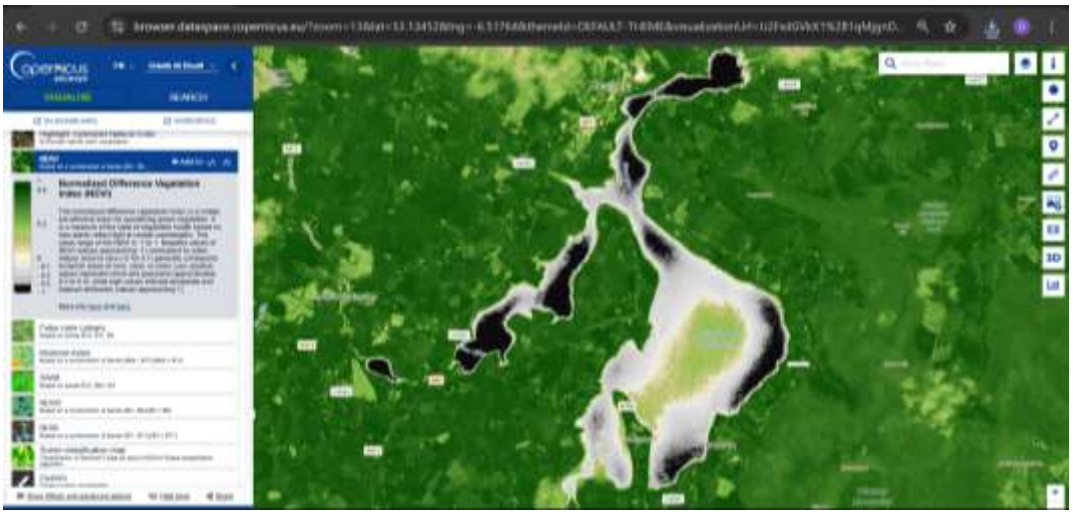


Figure 19. Region of Interest using the Copernicus Inter with NDVI filter during August 2025.
[https://browser.dataspace.copernicus.eu]

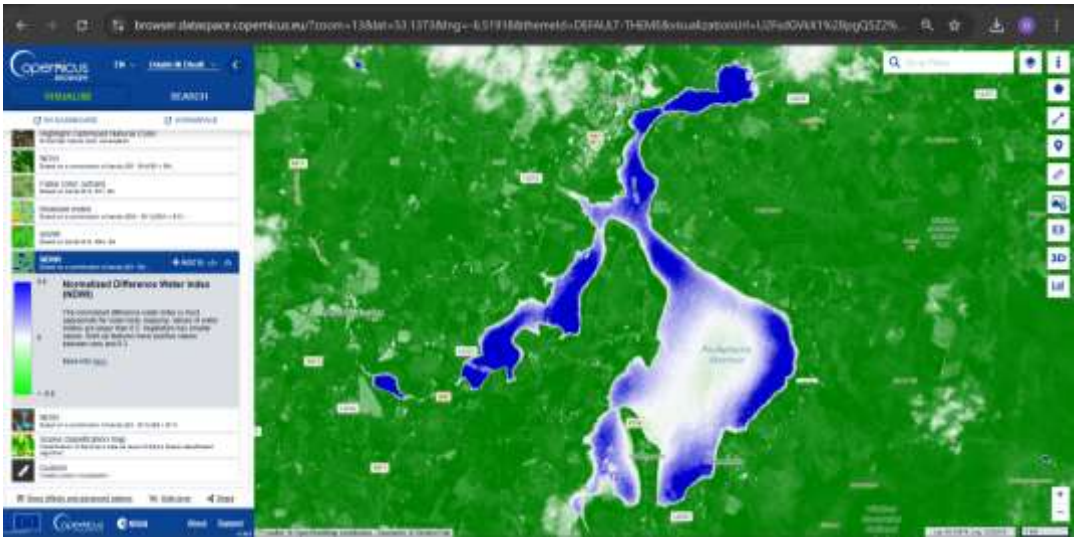


Figure 20. Region of Interest Analysis using the Copernicus Inerface with NDWI filter during August 2025
[https://browser.dataspace.copernicus.eu]

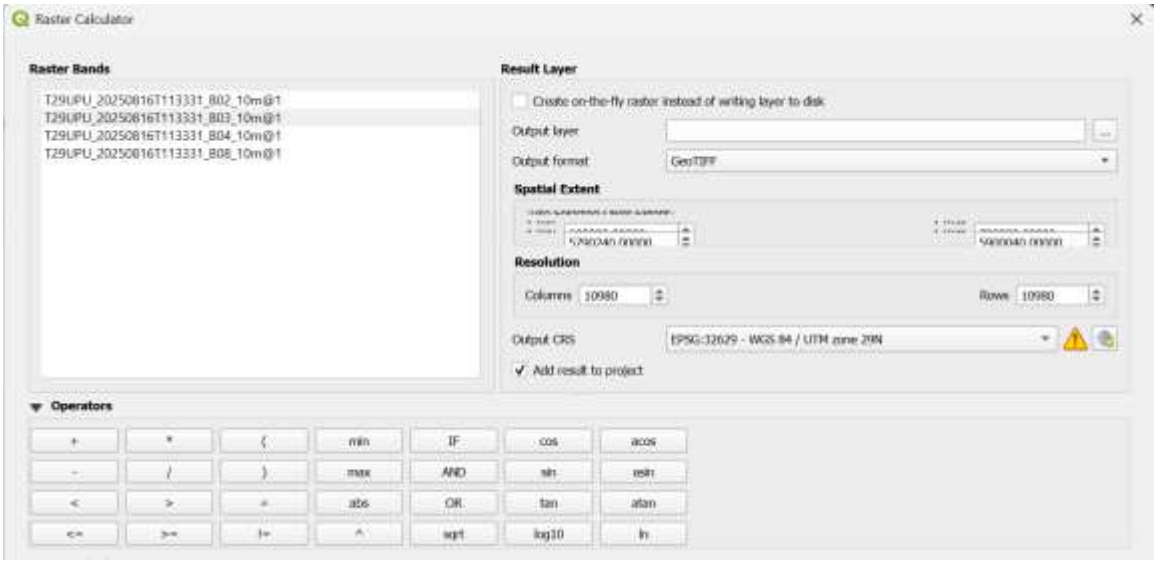


Figure 21. Raster calculation bands screenshot using QGIS application.

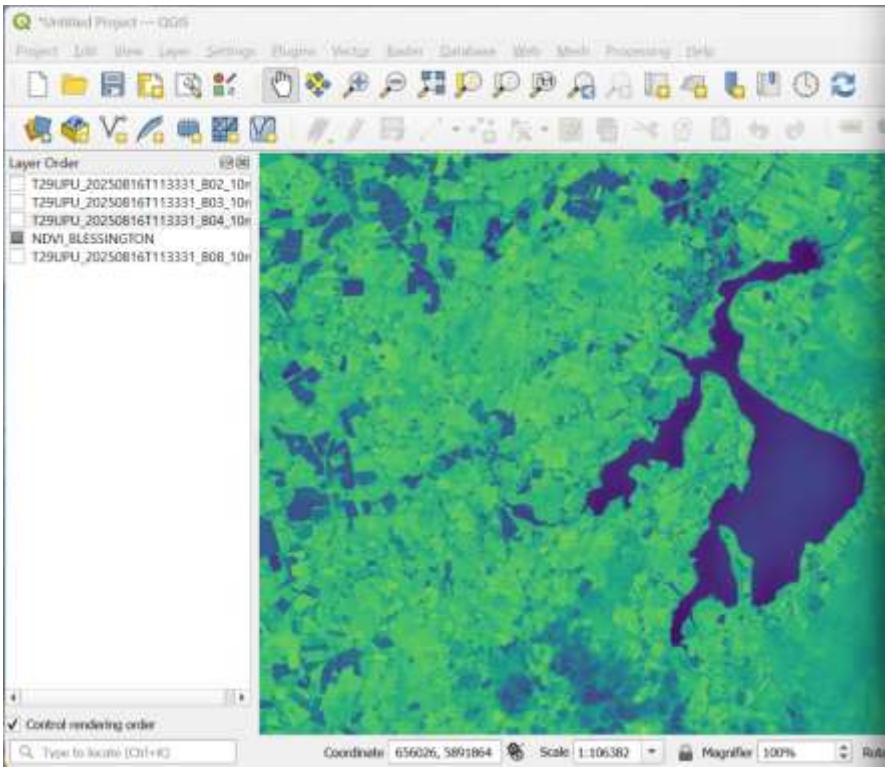


Figure 22. Sentinel-2 Calculated NDVI using masks on QGIS using raster bands above.

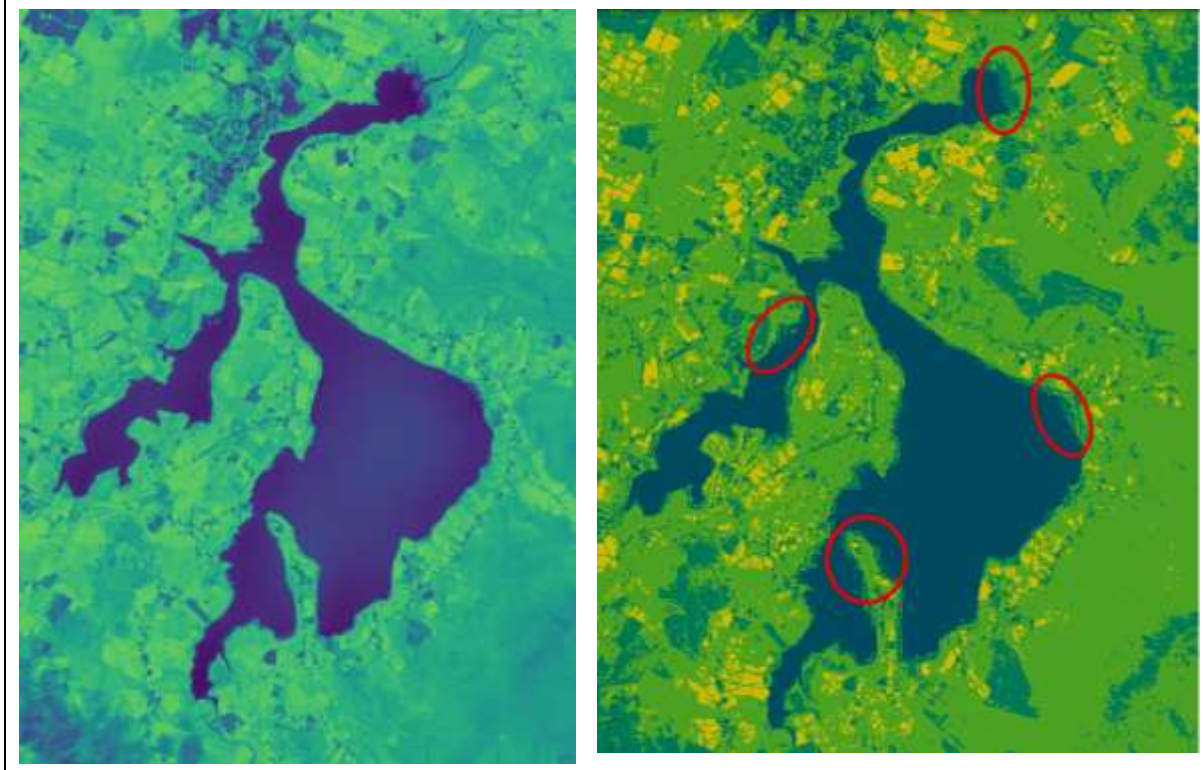


*Figure 23. Visual *possible* evidence of algal growth along shoreline from time series of NDVI with exaggerated filter use.*

Discussion

Overall, the results show a clear link between rainfall patterns and water-level changes in Poulaphouca Reservoir. The SWOT elevation data, where available, matched the dry and wet periods shown in the Met Éireann records, with lower levels during the warm, dry spells and rapid recovery once rainfall increased. The continuous 2025 SWOT record captured this especially well and showed how quickly the reservoir responds to changing weather. The Sentinel-2 work gave only limited evidence of algal activity. After checking NDVI and NDWI in the Copernicus viewer, only August 2025 could be analysed properly because 2024 scenes were too cloudy and I was also limited by time and storage. The NDVI results showed small patches of higher greening around shallow shoreline areas during the lowest water levels. These increases were minor and probably linked to exposed vegetation, floating organic material, or sediment rather than a full algal bloom. There was no sign of widespread or persistent algae in open-water areas. Taken together, the results suggest that while short-term shoreline greening can occur during dry and warm periods, conditions during the analysed months did not produce a strong or lake-wide algal event. More frequent satellite coverage and in-situ water-quality data would help confirm what exactly these small changes represent.

Conclusion

This study showed that satellite observations are effective for tracking both water level changes and potential algal activity in Poulaphouca Reservoir. The SWOT elevation record aligned well with local rainfall patterns, confirming that water levels fall during dry spells but recover quickly once rainfall returns. Even during periods of drought, the reservoir's elevation remained well managed, with no evidence of severe or prolonged

drawdown that would threaten supply. This supports the overall objective of assessing water availability for Dublin's drinking water system.

The Sentinel-2 analysis provided only limited evidence of algal presence. Minor increases in NDVI were detected along shallow shoreline areas during low-water periods, but these signals were small and likely related to exposed vegetation or sediment rather than any significant bloom. There was no indication of widespread or persistent algal growth in open-water areas. Based on these observations, algal activity during the analysed periods appears minimal, suggesting no clear risk to drinking water quality.

Overall, the findings indicate that Poulaphouca Reservoir remained both safe and reliable as a drinking water source throughout the study period. The combination of SWOT water level data, rainfall records, and Sentinel-2 imagery offered a useful overview of seasonal dynamics, and the approach provides a strong baseline for continued monitoring of water security and environmental conditions in one of Ireland's most important reservoirs.

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SWOT mission, hydrology products & data gaps

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Data availability

This study used publicly available satellite data from the European Space Agency (ESA) and NASA. Sentinel-1 C-band SAR and Sentinel-2 MSI imagery were obtained from the Copernicus Data Space Ecosystem. SWOT Water Surface Elevation (WSE) products were accessed through NASA's Physical Oceanography Distributed Active Archive Center (PO.DAAC). All data were processed using ESA SNAP 13.0, QGIS 3.34, and Python 3.11. The datasets are fully open-access and can be freely downloaded for research and educational use.

Appendix

Code used to successfully download SWOT data through Ubuntu Terminal.

```
# 1. Activate EZ-InSAR HydroOcean environment
conda activate EZ-InSAR-3

# 2. Create a working directory for Blessington SWOT data
mkdir -p ~/SWOT_Blessington
cd ~/SWOT_Blessington

# 3. DOWNLOAD SWOT L2 LAKE SP PRODUCTS FOR BLESSINGTON

ezinsar hydroocean download \
    -s SWOT \
    -b -6.618940,53.068340,-6.412946,53.210033 \
    --download \
```

```
    -u 'EarthData_Username' \ #This needs to be edited
    -p 'EarthData_Password' #This too

# 4. EXTRACT SWOT LAKE DATA TO GEOJSON
ezinsar hydroocean extractSWOT \
    -i ./ \
    -b -6.618940,53.068340,-6.412946,53.210033 \
    -o blessington_raw.geojson \
    -m Lake

# 5. FILTER BY THE ACTUAL BLESSINGTON LAKE ID

python3 << EOF
import geopandas as gpd

# Load extracted SWOT LakeSP GeoJSON
g = gpd.read_file('blessington_raw.geojson')

# Filter for the Blessington/Poulaphouca lake_id
g2 = g[g['lake_id'] == '2340000673']

# Save filtered lake-only dataset
g2.to_file('Blessington_Lake.geojson', driver='GeoJSON')

print("Filtered Blessington file written to Blessington_LakeID.geojson")
```

Field Title	Meaning
obs_id	Unique ID for this *specific observation* (one polygon at one time).
lake-id	ID for the lake in the SWOT global lake database. All polygons belonging to the same lake share the same lake_id.
reach_id	Usually for river reaches; often "no_data" for lakes (as in your case).
overlap	Number of overlapping passes or tiles contributing to this polygon. Higher values mean the area was captured in more SWOT swaths.
n_overlap	Number of *independent* swaths that observed the lake during that pass.
time	Observation time in TAI seconds (NASA's spacecraft time scale).
time_tai	Same idea; sometimes duplicated depending on product version.
time_str	Human-readable timestamp in UTC ("17/01/2024 13:17:09").
wse	Water Surface Elevation (metres). This is the main value you use.
wse_u	Uncertainty in WSE (1-sigma).
wse_r_u	Random uncertainty component.
wse_std	Standard deviation of elevation across the polygon. This is helpful for checking how noisy the measurement is.
area_total	Total area of water detected in this polygon (km^2).
area_tot_u	Uncertainty in area_total.
area_detect	Area that the satellite confidently detected (robust water).
area_det_u	Uncertainty in area_detct.
layovr_val	Fraction of the polygon potentially affected by radar layover (distortion near steep terrain). Values close to 1 mean low confidence.
width_dist, slope	Additional geometric metrics - usually not essential for lake-level time series work.

Figure 24. Index of Titles from SWOT database produced in QGIS during result acquisition.